



Artificial Intelligence and Machine Learning

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Department of Computer Applications

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AI Landscape

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What is Learnt So far?



How does the Learning Agent learn?

- **Rule-Based Learning**
- **Machine Learning**

Rule-Based Learning Agent:

System that makes decisions using predefined rules.

Structure:

IF condition → THEN action

Example:

IF temperature > 30°C → Turn ON fan

Used in

- Expert systems
- Early medical diagnosis systems
- Simple chatbots
- Thermostat system

Limitation

- **Knowledge Acquisition Problem:** Forget cases, incomplete knowledge
- **Scalability Problem:** managing rules when it increases exponentially
- **No Learning Ability:** Do not improve on experience
- **Uncertainty:** ambiguity, noise
- **Maintenance problem:** updating rules is difficult and is time consuming

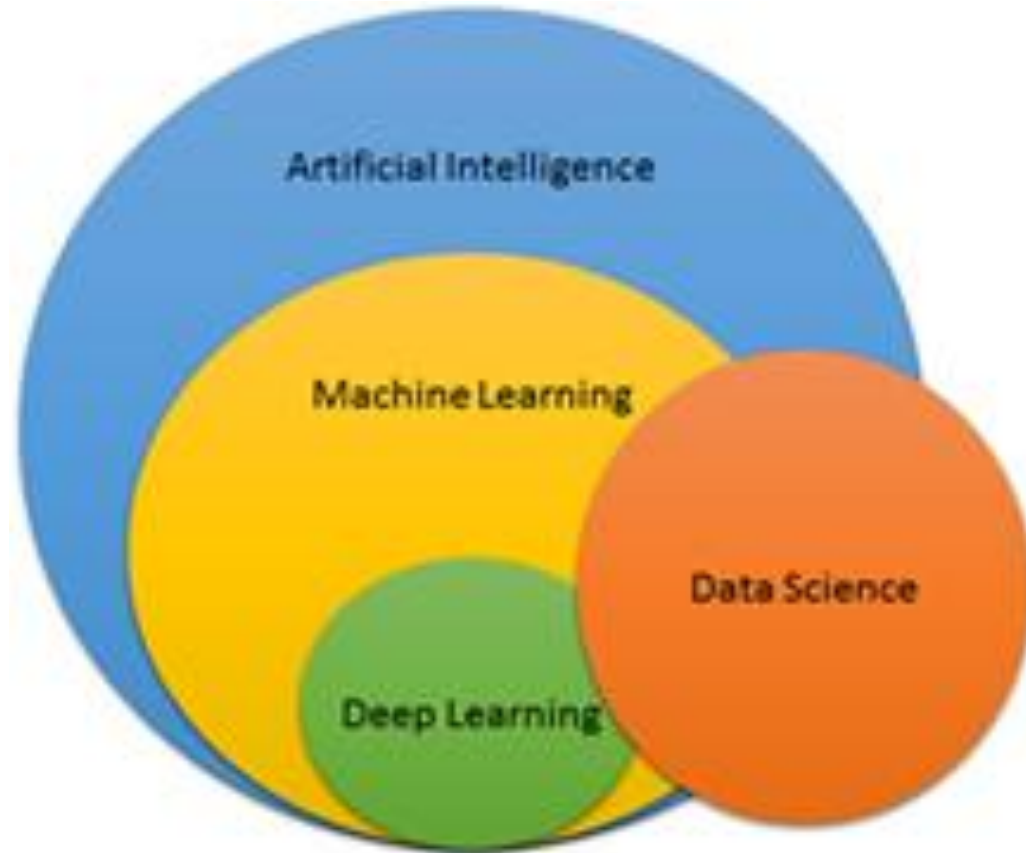
Hence Moving from **Rule Based to Machine Learning**

AIML

Learning Agent

Machine Learning

- Learns from data
- Adapts
- Improves over time



Machine Learning (ML) is a subset of AI

It provides us tools and techniques to explore, learn and analyze the data.

Deep Learning (DL) is further a subset of ML

The main idea behind it is to make machines learn by mimicking the human brain.

we create a neural network architecture to learn at micro level

Data Science (DS) is basically **drawing insights** from structured and unstructured data

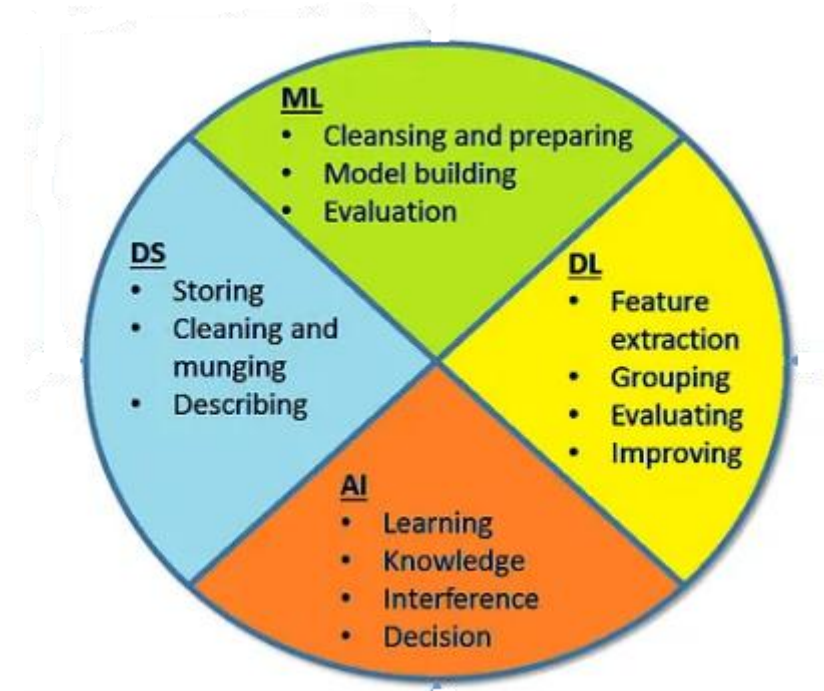
Use different **visualization tools, statistics, and probability** to gain these insights.

Artificial Intelligence- "Human Intelligence exhibited by machines".

Machine Learning - "An approach to achieve Artificial Intelligence focusses on algorithms learning from data"

Deep Learning - "A specialized technique for implementing machine learning using neural networks"

Data Science: "Art and Science of drawing actionable insights from the data".



Use Case: Uber Trips

AI : Uber AI powered systems for route optimization

ML: Dynamic pricing adjustments and predictive analytics

DL: Fraud detection capabilities (fake accounts, suspicious ride request)

DS: Drive insights (from which places the frequent trips are made)

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THANK YOU

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